Manual

Correction Functions for the PP-PDC Interfacing Module

SAP-PPPDK 3.0

Version 1.2.19800

Last changed on: 06.08.2020

Contents

1 Cancellation of PP Uploads

Summary

Use options

The SAP-PPPDCC function package extends the HYDRA interfacing module to SAP PP via PP-PDC by a cancellation option that is not provided for in the SAP PP-PDC interface.

Implementation notes

Use the function package to :

- use production orders in SAP PP and
- to transfer uploads from MES to SAP and
- to pass on corrections from MES to SAP to minimize maintenance works and to maintain the consistency of the confirmation data in both systems.

Integration

The function package uses the data entered into BDE.

The corrected data records are transferred via the PP_PDC interface.

Scope of functions

- Correction functions for PP-PDC link module
    - Functions to transfer corrections from HYDRA to SAP PP from Rel 4.6C.
    - Integration into the SAP business framework by synchronous RFC
    - Cancellation of the SAP uploads using SAP BAPI ProdOrdConfirmation.Cancel
    - Assurance of data consistency in the integration into the PP-PDC environment.
    - Notification in case that an upload cannot be canceled (condition: SAP-ESK license).

2 SAP BAPI

The notion Business Application Programming Interface (BAPI) SAP provides a series of predefined interfaces enabling partner systems (R/3, R/2 as well as third-party systems) to access the functions of the SAP R/3 and/or ECC system.

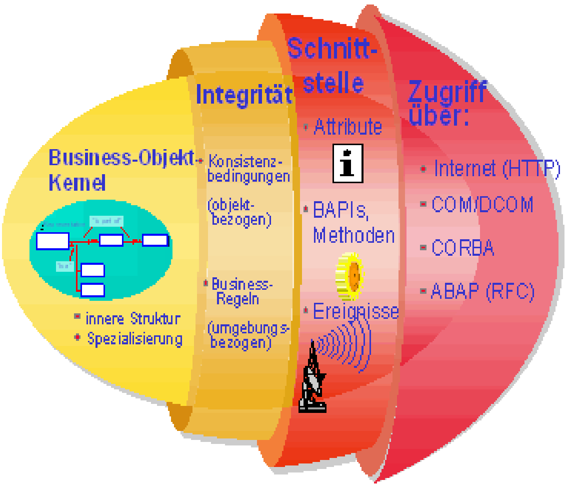
SAP defines BAPI as follows:

Standardized programming interface providing external access to processes and data of the SAP system.

Business Application Programming Interfaces (BAPIs) are defined as SAP business objects and/or SAP interface types in the Business Object Repository (BOR).

BAPIs provide an object-oriented view of the business components of the SAP system. They are implemented and stored as RFC-capable function modules in the Function Builder of the ABAP Workbench.“[1]

[1] Source. http://help.sap.com



Kernel:

Kernel that includes object data and the structure,

Integrity layer:

Business logic of the business object

Interface layer:

Provides for the communication with distributed systems

Access layer:

Defines the access technology, e.g. tRFC

Source:

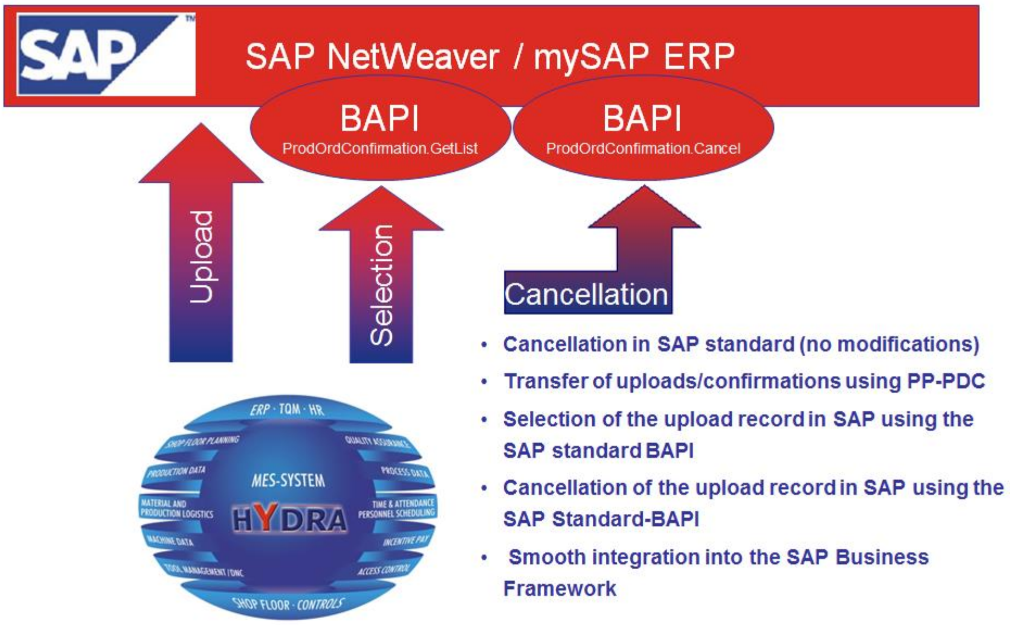
http://help.sap.com/saphelp_nw70/helpdata/de/5a/ccb4c5808311d396b40004ac96334b/frameset.htm

3 Cancellation Process

Technical basics of cancellation

Technically, uploads/confirmations are cancelled in SAP in two basic steps:

- Selecting the original data record to identify the SAP confirmation counter.
- Cancelling the original record using the confirmation counter.



The order key (order / sequence / operation / suboperation) and a unique ID (EX_IDENT) are required to select the confirmation counter. HYDRA generates this ID for each time ticket (original record) transferred to SAP. HYDRA sends this ID along with the time ticket. SAP keeps this ID with the operation confirmation/upload.

You can use the identified confirmation counter to clearly identify the time ticket (original record) and to cancel it in SAP.

Transferring cancellation records to SAP

In case of production orders, the postings recorded in HYDRA are promptly transferred to SAP and also posted there. If you change data subsequently in HYDRA, the same changes have to be repeated in SAP in order to avoid an imbalance between the data of both systems.

These are the steps required to execute and monitor the cancellation process:

Editing data in the HYDRA maintenance of postings

The following conditions must be met to transfer cancellations to SAP:

- The editing function "HYDRA maintenance of postings" is used.
- The order type is subject to confirmations/uploads.
- The order type allows for the posting to be changed once it has been uploaded.

The cancellation interface has not been released for use with the HYDRA event maintenance.

The system only generates cancellation records if the original data record has already been uploaded to SAP. The *Confirmed* column of the maintenance of postings indicates if a data record has already been uploaded to SAP.

If you change and save a data record in the maintenance of postings, only the current data record will be shown. The system still includes the original data record but does not show it.

Provision of data for the transfer to SAP

Time tickets are provided by converting the HYDRA data format into a meta format and transferring the data records to the HYDRA interface tables. The upload program generates a unique ID (EX_IDENT) during the transfer. This ID identifies each single record during the cancellation process.

If you use several HYDRA servers that upload data to an SAP instance, the EX_IDENT must differ from system to system to ensure uniqueness. To do so, the upload program (myerprck.exe/out) provides a parameter that guarantees this uniqueness. Use the following parameter to start the upload program:

/IDENT_PRAEFIX=XXX

Note: The prefix must **ONLY** include hexadecimal characters, i.e. 0-9 and A-F.

Displaying the records in MLE outbound transactions

After confirming the data records, they can be transferred to SAP in a meta format. The *Confirmed* column of the maintenance of postings indicates such data records using the icon ☑.

In the MLE outbound transactions (menu: File --> System information --> MLE communication --> Outbound transactions), you can identify the uploads to SAP via the PP-PDC interface and the cancellation records as follows:

	Time ticket uploads	Cancellation records
Message type	PPCC2PRETTICKET	E2BP_PP_TIMETICKET_C
IDoc type	PPCC2PRETTICKET01	PPCC2PRETTICKET01
Segment type	E2BP_PP_TIMETICKET	E2BP_PP_TIMETICKET_C

Segment E2BP_PP_TIMETICKET_C

The segment of cancellation records E2BP_PP_TIMETICKET_C is structured as follows:

Field name	T	L	D	Description	Usage in HYDRA
CONF_NO	NUMC	10	0	Upload number of the operation	Upload number of the operation
ORDERID	CHAR	12	0	Order	According to configuration (*1)
SEQUENCE	CHAR	6	0	Sequence	According to configuration (*1)
OPERATION	CHAR	4	0	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	0	Suboperation	According to configuration (*1)
PROC_START_TIME	TIME	6		Time when "starting processing"	Not used
EX_IDENT	CHAR	32	0	Unique reference to data record	Combined key: transfer parameter + ade_protokoll.verweis
POSTG_DATE	DATS	8	0	External date of entering the confirmation/upload	Depending on the program parameter /SDAT_STORNO of the program myerprck.exe/out : Not set (by default) : Change date of the log record /SDAT_STORNO set : Shift date of the log record
CONF_TEXT	CHAR	40		Confirmation/upload text for the cancellation record.	Reserved for future developments.

The documentation of the standard interface describes the structure of the standard records of the PP-PDC interface.

Process of transferring data to SAP

The SAP upload request controls the data transfer to SAP. The HYDRA MLE distribution model starts the HYDRA upload client. This upload client either transfers the time tickets to SAP using IDoc and tRFC or cancels the cancellation records via sRFC in SAP.

Initially, the upload client is called for the confirmation/upload of time tickets. The provision dates of the individual data records result in a sorting, which must be adhered to for serialization purposes.

In a first step, the upload client attempts to transfer time tickets to SAP. If this is possible, the upload client selects all time ticket records up to the first cancellation record and transfers the selected time tickets to SAP via IDoc. If it is not possible to select time tickets, the upload client checks whether there are cancellation records that have to be transferred to SAP.

If the upload client finds a cancellation record it will try to cancel it in SAP. Since cancellations are made using synchronous RFCs, you can state immediately whether a cancellation was successful.

If the cancellation process was successful the upload client determines the next records ready to be transferred (time tickets or cancellation records) based on the chronological order of their provision and transfers them to SAP.

If the cancellation was not successful, the cancellation record will be set to "TODO" in the interface. Consequently, the record is registered for the next transfer. In this confirmation/upload cycle, no additional data records will be transferred to SAP that were provided after this record.

The number of posting attempts is managed for the data record in order to prevent the confirmation/upload interface from being completely blocked for this message type. The upload client provides a parameter that allows to define after how many attempts a data record will be qualified as incorrect and when to proceed with the "normal" uploads.

The parameter "/REDO_CANCEL=" of the program hysapupl.exe is included in the HYDRA MLE distribution model and can be changed at any time.

The SAP upload request triggers the next transfer cycle, therefore this request should be transferred to HYDRA at intervals of 5 minutes.

By default, the upload client cancels one posting record in SAP per posting record in HYDRA. There might be specific, customer-related situations where the system generates several SAP posting records per HYDRA posting record.
In order to handle this situation, you can set the parameter "/MULTIPLE_CANCEL“ of the hysapupl.exe program in the HYDRA MLE distribution model. This will cancel all SAP posting records existing for a HYDRA posting record.

SAP operation status

HYDRA shows the same behavior when canceling data via the interface as SAP when collecting data via SAP-GUI. If a partial confirmation/upload is canceled for already completed operations and if this is entered again, SAP will set the operation status from finally confirmed to partially confirmed.

After correction of a "Posting/ Partial confirmation" (L20) in HYDRA, HYDRA will transfer the new "Posting/ Partial confirmation" also as record type L20 to SAP, irrespective of the operation status in HYDRA. If a final confirmation (L40) has been corrected in HYDRA, HYDRA will transfer this new final confirmation as record type L40 to SAP.

4 Manual Maintenance

Usage

If the automatic cancellation via the correction functions for the PP-PDC interfacing module was not successful, you will have to maintain manually.

Requirements:

Use the HYDRA interfacing module to SAP PP via PP-PDC and the correction functions for the PP-PDC interfacing module.

Approach

If a specific data record cannot automatically be canceled in SAP, it will be presented as incorrect in the MLE Outbound transactions (status "DONE ERROR" + red light). In this case, the data record must be canceled manually in SAP. To do so:

cancel the data records via the field "EX_IDENT" of the PP-PDC interface. Also the segment E2BP_PP_TIMETICKET contains this field for the data record to be canceled. The content of this data record will be displayed in the MLE Outbound transactions via the function "Show data segments for the transaction".

SAP will generate an internal number - the confirmation counter - for each confirmed time ticket. In the production order, the confirmation counter of the original record can be determined as follows:

SAP production order (CO02/ CO03) → transaction overview → transaction details → Transaction confirmations (Menu) → Transaction confirmations: Detail

In the "Administration" tab the field "Ext. Key" presenting the contents of the field EX_IDENT from the interface will be displayed. This field can be used to identify the data record and to determine the confirmation counter.

As soon as the confirmation counter has been determined like that, the original data record can be canceled using the SAP transaction "CO13". EX_IDENT can again be controlled in the presentation of this transaction.

Result

You will have proceeded manually to a cancellation that could not be made automatically.

5 Reaction to sRFC Errors

Usage

With sRFC-communication the calling system will receive next to a technical return code also business information regarding the success of an intended action.

The evaluation of the return parameters from the SAP function module can be optimized here as regards the question whether to reject specific data immediately (in this case the next data record would be edited) or whether to wait "online" for a couple of seconds before trying again to post the data to SAP.

Requirements

You use the synchronous RFC for the communication with SAP.

You execute the upload client hysapupl.exe/out by the command parameter /SINGLE_IDOC

Mapping

A general error handling function is integrated into the confirmation program to SAP hysapupl.exe/out that provides for a uniform error handling of sRFC-calls. This function assumes the error handling:

- if there is an error such as a RFC-exception and/or an exception of the function module,
- if the posting success is returned to a BAPIRET-structure,
- If the posting success is returned to a BAPIRET-table,
- if in case of the QM-IDI the posting success is returned to a QIERR-table.
- This uniform return code handling is made synchronously for all called RFC-modules, and in doing so existing island solutions were customized to this uniform processing method.

Processing is activated using the new program parameter/ RET_CODE_EVALUATION. This is used to evaluate the existing parameter/ REDO_CANCEL= for all synchronous calls. If this is not set, HKMPP-PDCC will be used as default. In addition, it will be checked in the sap_fb_return_cfg table whether it contains entries for the respective synchronous module.

In error handling the existing table sap_fb_return_cfg will be evaluated and any returned error will be handled correspondingly. To do so, the following logics is used:

PRIO 1 :

Error handling when the function module returns an exception.

This will be checked against the table sap_fb_return_cfg and be handled according to the configuration applying to that combination of function module name and exception.

If there is no entry, the data record will "normally" be handled as REDO-cancel and will automatically be set to "ToDo" and/or "DONE_ERROR".

PRIO 2 :

Error handling when the functional module does NOT return an exception but return the result in a return structure (Prio 2).

This will be checked against the table sap_fb_return_cfg and be handled according to the configuration applying to that combination of function module name and ret_type / ret_id / ret_number.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" DONE ERROR
- "A" DONE ERROR

PRIO3 :

Error handling when the functional module does NOT return an exception but return the result in a return table.

This return table may include more entries of different ret_type. This is the reason why this table will be used to iterate and in line with the following priority the table sap_fb_return_cfg will be searched for an entry:

- ret_type = A
- 2. ret_type = E
- 3. ret_type = W

If an entry will be found, the configuration in the table sap_fb_return_cfg will be used for further handling.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" (at least one entry – Prio 2) DONE ERROR
- "A" (at least one entry – Prio 1) DONE ERROR

PRIO4 :

Error handling when the functional module does NOT return an exception but return the result in a QUIERR table.

This return table may include more entries of different ret_type. This is the reason why this table will be used to iterate and in line with the following priority the table sap_fb_return_cfg will be searched for an entry:

- ret_type = A
- 2. ret_type = E
- 3. ret_type = W

If an entry will be found, the configuration in the table sap_fb_return_cfg will be used for further handling.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" (at least one entry – Prio 2) DONE ERROR
- "A" (at least one entry – Prio 1) DONE ERROR

If an entry is found in the table sap_fb_return_cfg for a return code, the system will interpret this entry as follows:

- ACTION = "A“ (only for SAP-PPPDCC)
  The data record will be marked immediately as DONE_ERROR without taking the program parameter "/REDO_CANCEL" into account.

- ACTION = "R"
  The system will wait for the time period specified in DELAY (in seconds) online (with live connection to SAP) and once this time period has elapsed will attempt to post the data once again.
  If this is possible (=successful), the data record will be marked as DONE.
  If this can again not be posted (e.g. because the data record is still being blocked in SAP), it will be marked in HYDRA as TODO (try again) or DONE_ERROR (impossible to post) while the value defined via the program parameter "/REDO_CANCEL" will be taken into account. If REDO_CANCEL is not explicitly specified during the program call, the DEFAULT value will be used.

Entries or changes to this table need to be agreed with MPDV since erroneous configurations may lead to unwanted side effects.

Table: sap_fb_return_cfg

Field	KEY	Type	L	D	Meaning	Notes/ Usage in HYDRA
FB_NAME	X	CHAR	30		Name of the functional module	Used to determine whether there are entries in this table to this FB
EXCEPTION	X	CHAR	30		Exception	Exception that is triggered by the module FUTURE USE
RET_TYPE	X	CHAR	1	0	Message type: S Success, E Error, W Warning, I Info, A Abort	Possible entries: "E" --> Error "A" --> Abort "W" --> Warning
RET_ID	X	CHAR	20	0	Messages, message class	Message type, e.g. "RU"
RET_NUMBER	X	NUMC	3	0	Messages, message number	Message number, e.g. "486"
RET_MESSAGE		CHAR	220	0	Message text	FUTURE USE
RET_LOG_NO		CHAR	20	0	Application log: Protocol number	FUTURE USE
RET_LOG_MSG_NO		NUMC	6	0	Application log : internal serial number of the message	FUTURE USE

Field	KEY	Type	L	D	Meaning	Notes/ Usage in HYDRA
RET_MESSAGE_V1		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V2		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V3		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V4		CHAR	50	0	Messages, message variable	FUTURE USE
RET_PARAMETER		CHAR	32	0	Name of the parameter	FUTURE USE
RET_ROW		INT4	10	0	Line in parameter	FUTURE USE
RET_FIELD		CHAR	30	0	Field in parameter	FUTURE USE
RET_SYSTEM		CHAR	10	0	System (logical system) that issued the message	FUTURE USE
ACTION		CHAR	2		Action to be executed	This defines which action will be executed by hysapupl.exe/out: "A" --> Cancel (immediate setting apart) "R" --> Online repetition
DELAY		NUMC	8		Delay in seconds	If "R"epeat is stored to the ACTION field, than this defines the time interval in seconds after which the repetition is to be made.